

Youssef Elebiary

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Summary

Computer science student at Cairo University with a strong foundation in AI, machine learning, and full-stack development. Experienced in building real-world applications using Python, C++, Flask, and modern ML frameworks. Certified by IBM, Google, and Huawei in AI and cybersecurity. Passionate about developing intelligent systems and seeking an internship or entry-level role in AI or software development to apply skills, learn, and contribute to impactful projects.

Experience

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| NLP Engineer Intern , Elevvo Pathways, Cairo | Jul 2025 – Aug 2025 |
| <ul style="list-style-type: none">• Fine-tuned a BART-base model for abstractive text summarization, achieving a ROUGE-1 score of 39.9 on the CNN-DailyMail benchmark.• Built end-to-end classification pipelines for Sentiment Analysis and Fake News Detection using TF-IDF and optimized standard classifiers.• Implemented unsupervised topic modeling using LDA and NMF, achieving a superior coherence score of 0.77 with NMF to extract latent themes from unstructured news data. | |
| Information Technology Intern , FPi-Future Pharmaceutical industries, Cairo | Aug 2025 – Sep 2025 |
| <ul style="list-style-type: none">• Enhanced a custom task management system by adding a "comment" feature that improved teamwork.• Contributed to an Order Receiving mobile app (Kotlin + Django API), partially integrated with the JD Edwards (JDE) ERP system.• Applied back-end development best practices while gaining hands-on experience with .NET and Angular. | |

Education

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| BSc in Computer Science , Cairo University | Sep 2023 – Jul 2027 |
| <ul style="list-style-type: none">• GPA: 3.6 | |

Certifications

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|---|----------|
| Microsoft Machine Learning Engineer , DEPI | Dec 2025 |
| IBM AI Engineering Specialization , IBM – Coursera | Sep 2024 |
| HCIA – Artificial Intelligence , Huawei ICT Academy | Sep 2024 |
| Object Oriented Programming Specialization , University of London – Coursera | Sep 2024 |
| Google AI Essentials , Google – Coursera | Sep 2024 |

Projects

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|--|---|
| RescuNet - AI Disaster Response Platform | github.com/YoussefElebiary/RescuNet |
| <ul style="list-style-type: none">• Developed a Graph Neural Network (GINEConv) to predict road traversability on highly imbalanced, sparse graphs and engineered a C++ routing engine to solve VRP in milliseconds.• Integrated YOLOv11 models for RGB hazard/human detection and Thermal human detection, alongside an NLP LSTM model to classify and prioritize distress signals.• Built a full React 19 website with MapLibre and FastAPI enabling users to visualize live drone feeds, perform real-time text analysis, and generate dynamic rescue routes. | |

Autonoma - Autonomous ML Pipeline

github.com/YoussefElebiary/Autonoma

- Automated the end-to-end machine learning lifecycle (EDA, preprocessing, and training) by engineering a **LangGraph** orchestration system that dynamically routes distilled state context, preventing local LLM VRAM exhaustion.
- Decoupled AI reasoning from code execution by integrating the **Model Context Protocol (FastMCP)**, securely executing generated JSON tool calls via **Polars** and **Scikit-Learn**.
- Designed a self-correcting Critic agent loop to autonomously audit preprocessing and modeling steps for data leakage and statistical fallacies prior to training and tuning predictive models.

AI Image Captioning

github.com/YoussefElebiary/VisionScribe

- Developed a captioning architecture using **InceptionV3** for feature extraction and **Bidirectional LSTMs** for sequence generation.
- Achieved a **BLEU-1 score of 0.54** and **BLEU-4 of 0.15**, validating the model's ability to generate semantically accurate descriptions.
- Implemented **Beam Search** decoding to improve caption quality and deployed the system via a **Flask** REST API with a web interface.

Fake News Classifier

github.com/YoussefElebiary/TruthGuard

- Engineered a dual-model classification system using **XGBoost** and **LightGBM**, achieving a **99.8% F1-Score** and **99.9% Precision** on the validation set.
- Developed a **Flask** web interface that allows users to toggle between high-precision and lightweight inference modes.

Image Editing Software

github.com/YoussefElebiary/PhotoWizard

- Built a desktop GUI app in **C++** using **Qt** to apply visual filters to images with real-time previews.
- Supported multiple image formats and optimized performance for smooth user interaction.

Skills

Technical Skills	Personal Skills	Languages
Python, C++ , HTML/CSS	Problem Solving	Arabic – Native
Scikit-learn, PyTorch	Communication	English – C1
Django, FastAPI	Teamwork	French – A2
Git	Time Management	German – A2

Extracurricular Activities

MSP Tech Club - Helwan University

Head of AI & Data Science

Jan 2026 – Present

- Leading the AI committee to design technical workshops and mentor junior members.
- Orchestrating the season plan to align with current industry trends in Data Science.

AI Committee Member

May 2025 – Dec 2025

- Collaborated on internal AI projects and organized community tech initiatives.